## Exercises

**Exercise 10.1 · Vocabulary.** A graph $G$ has vertices 1, 2, 3, 4, 5, 6 and edges 1-2, 1-3, 2-3, 3-4, 4-5, 4-6, 5-6.

1. Give the order of $G$ and the degree of each vertex, and check the property of the sum of the degrees.
2. Is $G$ complete? connected?
3. Give a walk of length 3 from 1 to 6 and a cycle through 5.

**Exercise 10.2 · Handshakes.**

1. Can one build a graph of order 7 all of whose vertices have degree 3?
2. Can six people each shake hands with exactly three others? If so, how many handshakes are there?

**Exercise 10.3 · A directed graph.** A directed graph has vertices $A$, $B$, $C$ and arcs $A \to B$, $A \to C$, $B \to C$ and $C \to A$.

1. Write its adjacency matrix $M$ (order $A$, $B$, $C$). Is it symmetric?
2. Compute $M^2$ and $M^3$.
3. How many paths of length 3 are there from $A$ to $A$? Write them out.

**Exercise 10.4 · A tournament.** Eight teams play a tournament: each meets each of the others once. Model with a graph and determine the number of matches.

**Exercise 10.5 · Degrees and the square of the matrix.** A graph has adjacency matrix

$$M = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}.$$

1. Draw it.
2. Compute $M^2$ and compare its diagonal with the degrees of the vertices.
3. Prove that for every undirected graph, $\left(M^2\right)_{ii}$ is the degree of vertex $i$.

**Exercise 10.6 · Testing connectedness.** A graph has vertices 1 to 5 and edges 1-2, 2-3, 3-1, 4-5. Let $M$ be its adjacency matrix.

1. Compute, with a calculator, $S = M + M^2 + M^3 + M^4$.
2. Is the graph connected? Justify directly, then check that this result can be read from the entries of $S$.

**Exercise 10.7 · Counting triangles.** A triangle of a graph is a set of three pairwise adjacent vertices.

1. Prove that the number of triangles is $\frac{1}{6}$ of the sum of the diagonal entries of $M^3$.
2. Apply this to the complete graph of order 4 and to the graph of exercise 10.5.